SAUMYA TIWARI

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SUMMARY

Final-year Computer Science Engineering student specializing in **backend** development and **scalable** web systems. Experienced in building **RESTful APIs** and data-driven applications using **Node.js**, Python, and Java. Strong foundations in databases, system design, and performance optimization with hands-on experience developing production-ready backend architectures.

TECHNICAL SKILLS

Backend & APIs: Node.js, Express.js, Flask, Spring Boot, RESTful APIs

Programming Languages: Java, C++, Python, JavaScript (ES6+)

Databases: SQL, MongoDB, Firebase

Frontend: React, Redux, Tailwind CSS, Bootstrap

Tools & Platforms: Git/GitHub, Docker, Linux, Vercel, Netlify, VS Code

Additional Skills: Data Analysis (Pandas, NumPy), OpenCV, Multithreading, Algorithm Optimization

EDUCATION

B.Tech in Computer Science & Engineering

AKS University, Satna (M.P.) | 2022–2026

CGPA: 7.72 / 10

Relevant Coursework: DSA, OS, DBMS, CN, AI/ML

INTERNSHIPS

Research Intern — Web Application & Data Processing

National Atmospheric Research Laboratory (ISRO) | Sept 2025 – Jan 2026

- Built **backend-driven** dashboards to visualize real-time lightning data using **Python and JavaScript**.
- Developed automated data processing scripts for atmospheric datasets, improving reporting efficiency and reliability.
- Collaborated with research scientists to **optimize backend data flow**, improving processing accuracy and system reliability.
- Focused on performance optimization and maintainability of web-based analytical tools.

Cyber Security Intern

Acmegrade | Feb 2024 – Apr 2024

- Performed **vulnerability assessment and penetration testing** on web applications following **OWASP Top 10** guidelines.
- Identified and mitigated security issues including **SQL injection, XSS, CSRF, and authentication flaws**, strengthening backend security.
- Analyzed network traffic and application behavior using industry tools to assess and improve system defenses.

SELF CONCEPTUALISED PROJECTS

Aura — AI Voice Assistant (MERN Stack Application)

MongoDB, Express.js, React, Node.js, Web Speech API | Oct – Dec 2025

- Developed a full-stack AI voice assistant enabling **real-time voice** interaction using the **Web Speech API**.
- Designed and implemented **RESTful APIs** with **Node.js** and **Express** to process commands and manage user sessions.
- Built **backend services** for speech processing and stored interaction history using **MongoDB**.
- Implemented asynchronous request handling and **modular backend** architecture to improve scalability and performance.
- Implemented **asynchronous request handling** to process voice commands efficiently, **reducing** command response latency by **~30%.**

VastraVista — AI-Based Fashion Recommendation System

Python, OpenCV, React, Tailwind CSS | May – Sept 2025

- Built an AI-driven web application that analyzes user skin tone using **computer vision (OpenCV)** and recommends complementary clothing colors.
- Designed a responsive frontend with **React and Tailwind CSS** and integrated a lightweight ML pipeline for real-time recommendations.
- Focused Implemented end-to-end system integration focusing on performance and accuracy.

AI-Powered Notes & Task Manager

React, OpenAI API | June – July 2025

- Developed a productivity web application that generates intelligent task and note suggestions using the **OpenAI API**.
- Improved application performance by **~30%** through local storage caching and asynchronous data handling.
- Implemented clean state management and scalable component design.

Job Finder Web Application

React, Tailwind CSS, REST APIs | Jan – Apr 2025

- Built a dynamic job search platform with **API integration**, filtering, and saved search functionality.
- Deployed on **Vercel**, ensuring fast response times and reliable access across devices.
- Emphasized modular frontend architecture and maintainable API consumption.

AI-Based Chess Game — Interactive Web App

JavaScript, HTML, CSS | Aug – Sept 2025

- Designed and deployed a real-time chess game featuring an **AI opponent** with custom move-prediction logic.
- Implemented lightweight algorithms for decision-making to ensure fast response and smooth gameplay.
- Deployed on **Vercel** with a responsive UI for cross-device compatibility.

HACKATHON

IBM National Hackathon — Airline Reservation System with Dynamic Pricing

Bengaluru | Aug 2025

- Designed and implemented a **multi-threaded airline reservation system** using **C++**, supporting concurrent bookings and dynamic surge pricing.
- Built core modules for **seat allocation, pricing logic, and loyalty programs**, optimizing data handling and system throughput.
- Improved booking efficiency by **~25%** through concurrency control and optimized algorithms.
- Led a **4-member team** in a 24-hour hackathon, coordinating system design, task distribution, and final integration.

ACHIEVEMENTS & LEADERSHIP

- National Hackathon Finalist (Top 1%) — IBM National Hackathon 2025.
- Led a 4-member team in a 24-hour hackathon, owning system design, task breakdown, and final integration.
- Mentored 100+ students in Data Structures & Algorithms, focusing on problem-solving and coding efficiency.